End-to-End Flow Test Report(Model 1)

Data Source

- **Model Inputs:** Derived from typical DC and AC power usage scenarios for SMEs, based on historical data and estimates from the **SME Energy Consumption dataset** and **NASA POWER data** previously used in Model 1.
- Values for the 4 scenarios were chosen to reflect realistic ranges for **small/medium retail** and **small/medium manufacturing** operations.

Tested Scenarios

Scenario	Business Type	Inputs (DC / AC Power)	Predicted Output (kWh/day)	Observations / Recommendations
1	Small Retail Store	DC=200, AC=150	950 kWh	Output is reasonable for a small retail store. Consider rounding predictions to the nearest kWh for dashboard clarity.
2	Medium Retail Store	DC=500, AC=450	2600 kWh	Output aligns with expected consumption. No major issues observed.
3	Small Manufacturing	DC=800, AC=700	3900 kWh	Output is slightly high but realistic for light manufacturing equipment. Suggest checking peak usage assumptions in the dashboard.
4	Medium Manufacturing	DC=1500, AC=1200	7100 kWh	High energy usage is expected. Recommend dashboard shows cost comparison vs projected solar cost for transparency.

General Observations

- All outputs are **within realistic ranges** for the tested business types.
- The system handled inputs smoothly; no missing input errors occurred.
- Predictions match expectations based on business type and size.